DRUGS · ANTIBIOTICS: CEPHALOSPORINS AND CARBAPENEMS

Meropenem

Meropenem is a carbapenem: the broadest beta-lactam in routine use, active against almost all Gram-negative bacilli including ESBL producers and Pseudomonas, most Gram-positives apart from MRSA, and anaerobes. In ENT it is reserved for resistant organisms and the critically ill: severe deep neck and mediastinal sepsis, hospital-acquired infection after major head and neck surgery, intracranial infection with resistant organisms, and necrotising otitis externa that has failed first-line treatment.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

MEROPENEM

	SUMMARY
Class	Carbapenem (beta-lactam)
How it works	Cell wall synthesis inhibitor, stable to almost all beta-lactamases including ESBL and AmpC
Spectrum	Enterobacterales including ESBL producers, Pseudomonas, Acinetobacter (some), streptococci, methicillin-sensitive S. aureus, anaerobes; not MRSA, not Enterococcus faecium, not Stenotrophomonas, not carbapenemase producers
ENT uses	Severe or hospital-acquired head and neck sepsis; descending mediastinitis; ESBL infection; meningitis and brain abscess with resistant Gram-negatives; failed treatment of skull base osteomyelitis
Adult dose	1 g every 8 hours intravenously; 2 g every 8 hours for meningitis and severe infection
Child dose	10 to 20 mg/kg every 8 hours; 40 mg/kg every 8 hours for meningitis (maximum 2 g per dose)
Avoid	Beta-lactam anaphylaxis (cross-reactivity with penicillin is low, about 1%)

	SUMMARY
Pregnancy	Use if needed

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

Carbapenems bind multiple penicillin-binding proteins and resist hydrolysis by nearly all serine beta-lactamases. Resistance comes from **carbapenemases** (KPC, OXA-48, NDM) and porin loss, and it spreads; every unnecessary carbapenem prescription adds to the problem, which is why most hospitals restrict them to microbiology approval.

2.2 Handling

Intravenous; renally excreted, so reduce the dose when the creatinine clearance is below 50 ml/min. It crosses the inflamed meninges and has a lower seizure risk than imipenem.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Severe neck and mediastinal sepsis

Descending necrotising mediastinitis from a dental, pharyngeal or deep neck source carries a high mortality: secure the airway, drain the neck and mediastinum (with thoracic surgeons), and give broad cover such as meropenem with vancomycin or linezolid, narrowed by cultures.

3.2 Resistant and hospital organisms

After major resection and free-flap reconstruction, fistula and wound infection may involve ESBL Enterobacterales or resistant Pseudomonas: meropenem is used when cultures demand it, not as a reflex.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Diarrhoea, **C. difficile** and candidiasis; selection of resistant organisms.
- Seizures (less than imipenem), particularly in renal failure or CNS disease.
- Hypersensitivity, raised liver enzymes, thrombocytopenia.

4.2 Interactions

Sodium valproate: carbapenems lower valproate levels profoundly within a day, and seizures follow; the combination should be avoided.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- What carbapenems cover and what they do not (**MRSA**, E. faecium, Stenotrophomonas).
- The **valproate** interaction.

- Antimicrobial stewardship: microbiology advice, cultures first, review at 48 to 72 hours.

5.2 Common pitfalls

COMMON PITFALLS

- Using meropenem first-line for community-acquired head and neck infection.
- Forgetting MRSA cover in the septic post-operative patient.
- Giving it to a patient taking valproate.

Written for this app from BNF and BNF for Children; NICE and specialist society guidance; Scott-Brown; Cummings. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.